Compact ordered spaces and the ball fixed point property for $C(K,p)$

arXiv automatic math run `fa_banach_001`

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Source Question

The source paper is Aviles–Japon–Lennard–Martinez-Cervantes–Stawski, *The ball fixed point property in spaces of continuous functions*, arXiv:2506.17995 [1]. It asks in final Question (i) whether there exists an infinite compact space K and a non-isolated point $p \in K$ such that

$$C(K,p) = \{f \in C(K) : f(p) = 0\}$$

has the ball fixed point property BFPP.

> remainder $\mathbb{N}^*$ is a much more delicate set-theoretic question. One can construct such P-points under the Continuum Hypothesis [29], but there are other models of set theory in which $\mathbb{N}^*$ contains no P-points [35].
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> We finally conclude the paper by highlighting several questions that may arise after reading the article, which we believe could be of significant interest for future research in the area:
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> i) Is it possible to find an infinite compact set K and a non-isolated point $p \in K$ such that $C(K,p) = \{f \in C(K) : f(p) = 0\}$ has the BFPP?
>
> ii) Is it possible to construct an explicit example of a fixed point free nonexpansive mapping from the unit ball of ℓ_∞ / c_0 into itself (without turning to CH)?
>
> iii) Is it possible to exhibit an example of a fixed point free nonexpansive mapping from the unit ball of $C(K)$ into itself, whenever $C(K)$ fails to be order-complete? In this case, we would have a topological characterization of the BFPP for Banach spaces of continuous functions: $C(K)$ satisfies the BFPP if and only if K is extremally disconnected.
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> References

The source paper proves failure when p is a non-P-point. Previous lane-15 packets proved failure for ordinal compacta and, more generally, for points admitting a clopen ordinal tower. This packet gives a different extension: all compact linearly ordered spaces are negative examples.

Idea of the Proof

Let K be a compact linearly ordered topological space and let p be non-isolated. From one side of p we can choose a closed subspace $F \subset K$ homeomorphic to an ordinal interval $[0, \kappa]$, with κ

a nonzero limit ordinal and p corresponding to κ. The old ordinal shift gives a fixed-point-free nonexpansive self-map of the unit ball of $C(F,p)$.

The point of the ordered setting is that restriction to a closed ordered subspace has a norm-one linear right inverse. On each convex component of $K \setminus F$, extend a function on F by keeping it constant if the component has only one boundary point in F, and by interpolating between the two boundary values if it has two. This positive extension operator has norm one and preserves the condition of vanishing at p. Therefore a bad map on $C(F,p)$ transfers to a bad map on $C(K,p)$.

Ordered Extension Lemma

Lemma 1. *Let K be a compact linearly ordered topological space and let $F \subset K$ be closed. Then there is a positive linear operator*

$$E : C(F) \longrightarrow C(K)$$

with $\|E\| = 1$ and $Ef|_F = f$ for every $f \in C(F)$. Consequently, if $p \in F$, then E restricts to a norm-one linear right inverse

$$E : C(F,p) \longrightarrow C(K,p)$$

for the restriction map $R : C(K,p) \to C(F,p)$.

Proof. The components of $K \setminus F$ are convex open intervals in the order. For such a component I, the boundary $\overline{I} \setminus I$ is a nonempty subset of F with either one or two points. If it has one point a, put $(Ef)(x) = f(a)$ for $x \in I$. If it has two points $a < b$, choose, by normality of the compact interval $[a,b]$, a continuous function $\theta_I : [a,b] \to [0,1]$ such that $\theta_I(a) = 0$ and $\theta_I(b) = 1$, and put

$$(Ef)(x) = (1 - \theta_I(x))f(a) + \theta_I(x)f(b), \qquad x \in I.$$

Finally set $(Ef)(x) = f(x)$ for $x \in F$.

Linearity and positivity are immediate, and $\|Ef\|_\infty \le \|f\|_\infty$ because every value on a gap is a convex combination of boundary values. Since E fixes F, $\|E\| = 1$ when $F \neq \emptyset$.

It remains only to note continuity at points of F. Let $y \in F$ and let $x_i \to y$ be a net in K. If $x_i \in F$ eventually, continuity follows from that of f. Otherwise pass to points lying in components I_i of $K \setminus F$. If a component I_i is adjacent to y, then the chosen interpolation tends to the boundary value $f(y)$ as $x_i \to y$. If I_i is not adjacent to y, then its boundary points in F eventually lie in every prescribed order-neighborhood of y; otherwise convexity would force the component to cross a fixed point of F near y. The continuity of f on F then makes all boundary values, and hence their convex combinations, converge to $f(y)$. Thus $Ef \in C(K)$.

If $p \in F$ and $f(p) = 0$, then $Ef(p) = f(p) = 0$, so the same operator maps $C(F,p)$ into $C(K,p)$. It is a right inverse to restriction because it fixes every point of F. $\square$

Ordinal Skeleton Lemma

Lemma 2. *Let K be a compact linearly ordered topological space and let $p \in K$ be non-isolated. Then there are a nonzero limit ordinal κ and a closed $F \subset K$ such that $p \in F$ and F is homeomorphic to $[0,\kappa]$, with p corresponding to κ.*

Proof. Since p is non-isolated, p is approached from at least one side. We write the proof for the case where p is in the closure of $\{x \in K : x < p\}$; the right-hand case is obtained by reversing the order.

Let $\kappa = \mathrm{cf}(\{x \in K : x < p\})$, the cofinality of the initial segment below p. Then κ is a nonzero limit ordinal. Choose an increasing cofinal family $(y_\alpha)_{\alpha<\kappa}$ below p. By transfinite recursion, replacing it by a cofinal subsequence if necessary, choose a strictly increasing continuous family $(x_\alpha)_{\alpha<\kappa}$ such that

$$x_\lambda = \sup_{\alpha<\lambda} x_\alpha$$

for every nonzero limit $\lambda < \kappa$, and $\sup_{\alpha<\kappa} x_\alpha = p$. At a limit stage $\lambda < \kappa$, the supremum of the previous points is still below p; otherwise the initial segment below p would have cofinality at most $\lambda < \kappa$.

Set

$$F = \{x_\alpha : \alpha < \kappa\} \cup \{p\}.$$

The map $x_\alpha \mapsto \alpha$, $p \mapsto \kappa$, is a homeomorphism from F onto $[0, \kappa]$: continuity at nonzero limit stages is exactly the definition of x_λ, and neighborhoods of p contain tails of the cofinal family. The same description shows that F contains all its order limit points in K, hence F is closed in K. □

Partial Result

Lemma 3. *Let κ be a nonzero limit ordinal. Then*

$$C([0,\kappa],\kappa) = \{f \in C([0,\kappa]) : f(\kappa) = 0\}$$

fails BFPP.

Proof. Let $X = C([0,\kappa],\kappa)$. Define $S : B_X \to B_X$ by

$$(Sf)(\alpha) = \begin{cases} 1, & \alpha = 0, \\ f(\beta), & \alpha = \beta + 1 < \kappa, \\ f(\alpha), & 0 < \alpha < \kappa \text{ is a limit ordinal,} \\ 0, & \alpha = \kappa. \end{cases}$$

The same successor-shift continuity check used for ordinal compacta shows that $Sf \in X$: at every limit ordinal the shifted successor values converge to the already prescribed limit value of f, and at κ they converge to $f(\kappa) = 0$. The map is nonexpansive because each coordinate of $Sf - Sg$ is either 0 or a coordinate of $f - g$.

If $f = Sf$, then $f(0) = 1$, and transfinite induction gives $f(\alpha) = 1$ for every $\alpha < \kappa$: the successor step follows from $f(\beta + 1) = f(\beta)$, and the limit step follows from continuity. This contradicts $f(\kappa) = 0$. Hence S is fixed-point-free. □

Theorem 1. *Let K be an infinite compact linearly ordered topological space and let $p \in K$ be non-isolated. Then $C(K, p)$ fails the ball fixed point property.*

Proof. By Lemma 2, choose a closed $F \subset K$, a nonzero limit ordinal κ, and a homeomorphism $F \cong [0, \kappa]$ carrying p to κ. By Lemma 3, $C(F, p)$ fails BFPP. Equivalently, there is a fixed-point-free nonexpansive map

$$S : B_{C(F,p)} \longrightarrow B_{C(F,p)}.$$

Let $R : C(K, p) \to C(F, p)$ be restriction. By Lemma 1, there is a norm-one linear operator $E : C(F, p) \to C(K, p)$ with RE equal to the identity on $C(F, p)$. Define

$$T : B_{C(K,p)} \longrightarrow B_{C(K,p)}, \qquad T(f) = E(S(Rf)).$$

Since R, S, and E are all nonexpansive on the relevant unit balls, T is nonexpansive and maps the unit ball of $C(K,p)$ into itself.

If $f \in B_{C(K,p)}$ were a fixed point of T, then applying R would give

$$Rf = RESRf = S(Rf),$$

contradicting that S is fixed-point-free. Hence T is a fixed-point-free nonexpansive self-map of the closed unit ball of $C(K,p)$, so $C(K,p)$ fails BFPP. □

Scope and Limitations

This result extends the ordinal compact packet to all compact ordered spaces. It includes connected compact intervals and long-line type compacta, where a clopen-tower argument is not available. The result is still a partial answer to the source question: arbitrary compact Hausdorff spaces need not contain a closed one-sided ordinal skeleton with a norm-one extension operator.

Novelty and Verification Notes

The local run indexes were searched for arXiv `2506.17995`, BFPP, $C(K,p)$, ordered compacta, LOTS, and extension-operator language. No prior packet for compact linearly ordered spaces was found. A bounded web search on 2026-06-15 for exact combinations of $C(K,p)$, BFPP, and linearly ordered compact spaces found no later explicit answer to this subcase.

Human review should focus on Lemma 1, especially continuity of the interpolation extension at points of F which are limits of many convex gaps. Once that lemma is accepted, the transfer of the ordinal fixed-point-free map is formal.

References

[1] A. Aviles, M. Japon, C. Lennard, G. Martinez-Cervantes, A. Stawski, *The ball fixed point property in spaces of continuous functions*, arXiv:2506.17995, 2025.